ONE VANISHING EXT GROUP KILLS ALL HIGHER ONES

THE STACKS PROJECT AUTHORS

Editorial note. Counted span is the author's own proof environment, source lines 9206-9225, following the statement at 9198-9204.

Editorial note. No omissions: both extensions E' and E" are displayed and the splice composition identified.

Allowed standard prerequisites (editorial).

- Yoneda extension description of Ext. lemma-yoneda-extension (same chapter, cited at ~9209; cited for the existence half, whose proof is present in the chapter). Every class in Ext^i is delta of a Yoneda extension.

Source reference. Original lemma 06XU: sources/stacks-derived/pinned-original-source.tex, source lines 9048–9048. Navigation pointer only: the counted proof cites this statement by \ref as a standing input; the deduction the counted proof has to make is carried out inside the counted span, and no part of that deduction is delegated to this pointer.

Lemma 0.1. *Let $\mathcal{A}$ be an abelian category and let $p \geq 0$. If $\mathrm{Ext}^p_{\mathcal{A}}(B, A) = 0$ for any pair of objects A, B of $\mathcal{A}$, then $\mathrm{Ext}^i_{\mathcal{A}}(B, A) = 0$ for $i \geq p$ and any pair of objects A, B of $\mathcal{A}$.*

Proof. For $i > p$ write any class ξ as $\delta(E)$ where E is a Yoneda extension

$$E : 0 \to A \to Z_{i-1} \to Z_{i-2} \to \ldots \to Z_0 \to B \to 0$$

This is possible by Lemma 06XU. Set $C = \mathrm{Ker}(Z_{p-1} \to Z_{p-2}) = \mathrm{Im}(Z_p \to Z_{p-1})$. Then $\delta(E)$ is the composition of $\delta(E')$ and $\delta(E'')$ where

$$E' : 0 \to C \to Z_{p-1} \to \ldots \to Z_0 \to B \to 0$$

and

$$E'' : 0 \to A \to Z_{i-1} \to Z_{i-2} \to \ldots \to Z_p \to C \to 0$$

Since $\delta(E') \in \mathrm{Ext}^p_{\mathcal{A}}(B, C) = 0$ we conclude. □

Date: stacks-project:derived.tex:local-snapshot-20261012; GNU FDL 1.2 or later.